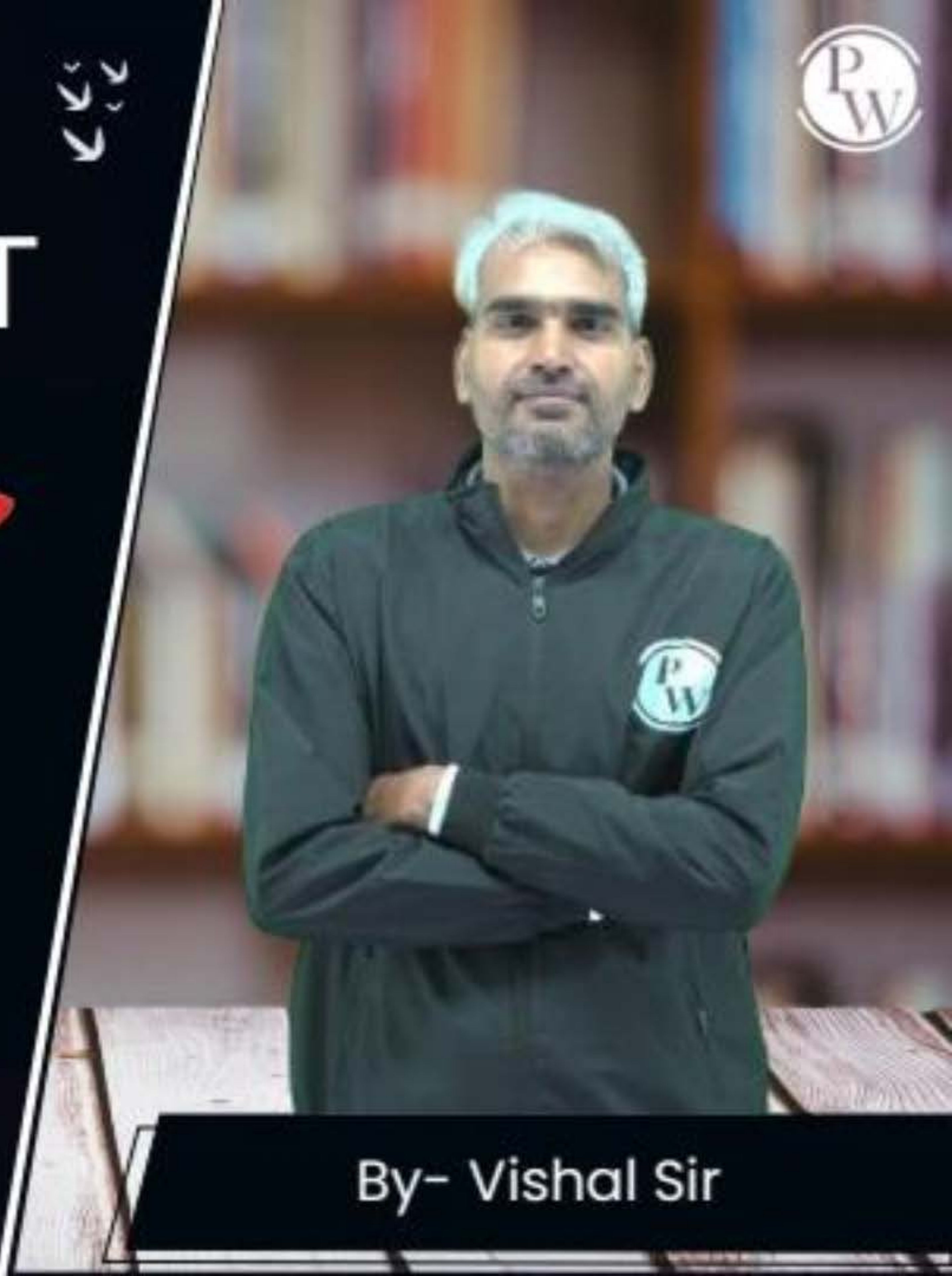
Computer Science & IT

Discrete Mathematics

Set Theory & Algebra

Lecture No. 01



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Topics to be Covered

- **Topic** Sets and representation of sets
- **Topic** Different types of sets
- **Topic** Number of subsets of a given set
- **Topic** Power sets

Topic : Syllabus

1. Set Theory & Algebra
   - Sets & Relations
   - Functions
   - Group Theory
2. Graph Theory
3. Mathematical Logic
   - Propositional logic
   - Predicate Logic
4. Combinatorics
   - Recurrence relation
   - Generating function

Slide

Topic : Set

A well-defined unordered collection of distinct elements is called as set.

eg. $A = \{1, 3, 5, 6, 8\}$

$B = \{1, a, b, \text{Jan}, \text{Feb}, \text{Mon}, \text{Ram}\}$

$$\boxed{\{1, 3, 7, 9, a, b\} = \{1, a, 3, b, 7, 9\}}$$

$\{1, 1, 3, 4, 4, 5\} \Rightarrow$ Actual Set will be $\Rightarrow \{1, 3, 4, 5\}$

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Topic : Representation of sets

There are three different ways in which set may be represented.

1. Roaster form or Tabular form:
2. Set-builder form:
3. Statement form:

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Topic : Roaster form or Tabular form

In this notation we will list all the elements of the set within the curly braces.

$A = \{1, 2, 3, 4, 5\}$

$B = \{1, 2, 3, \text{Jan}, \text{feb}\}$

$C = \{1, 2, 3, 4, \ldots\ldots\ldots\}$ All natural numbers.

$X = \{0, 7, 8, 13, 16, 21, 73, 75\}$

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Topic : Set-builder form

- In this notation we define the property and all the elements satisfying that property will be members of the set

- $A = \{x \mid \underbrace{x \in N \text{ and } x < 6}_{\text{Property}}\} = \{1, 2, 3, 4, 5\}$

Set Containing x such that

$D = \{y \mid y \in R \text{ and } 0 \le y \le 1\}$

We can not list the elements of set D
∴ it can not be represented using Roaster form

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Topic : Statement form

A = Set of all natural numbers less than or equal to 5.

D = Set of all real numbers between 0 & 1

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Topic : Cardinality of a set

{ Cardinality is generally defined for finite sets }

Cardinality of a set A is denoted by $|A|$ and $|A|$ represent the number of elements in set A.

eg $A = \{1, 2, 3, \text{Jan}, \text{Feb}\}$

$|A| = 5$

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Topic : Types of sets

Empty Set: A set with no element in it is called an Empty set.

Empty set can be denoted by $\{\ \}$ or ϕ

set of No member

$$\boxed{\{\ \} = \phi}$$

$$|\{\ \}| = |\phi| = 0$$

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$\{\ \} = \emptyset$

Empty set

$\emptyset \neq \{\emptyset\}$

$\{\emptyset\} = \{\{\ \}\}$

set

Element of the set, Here element of set is an Empty set

it is a set containing one element

Topic : Types of sets

Singleton Set: A set with exactly '1' element in it is called a singleton set

eg $A = \{a\}$

$B = \{1\}$

$C = \{ \underbrace{\{1, \{2,3\}\}}_{\text{it is one element}} \}$

set

Topic : Types of sets

Finite Set: A set that contains finite number of elements is called a finite set

eg: $\{1\}$, $\{1, 2\}$, $\{a, b, c\}$ etc..

$\{\ \}$ ← Empty set is also a finite set

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Topic : Types of sets

Infinite Set: A set that contains infinite number of elements in it is called an infinite set.

eg: A = set of all natural numbers

B = set of all real numbers

C = set of all Even integers

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Another Classification w.r.t. number of elements in the set

- Finite sets
  - Finite sets are always countable (or Countably finite)
- Infinite sets
  - Countably infinite sets
    - eg: A = Set of all Natural numbers
  - Uncountably infinite sets
    - eg. Set of all real no.s b/w 0 & 1

Topic : Types of sets

Equal Sets: To sets A & B are said to be equal if and only if Every element of set A is a member of set B and every element of set B is a member of set A.

eg $A = \{1, 2, a, b, c\}$ & $B = \{1, a, b, 2, c\}$

Here $A = B$

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eg

$A = \{1, 2, 3, 4\}$

$B = \{1, 2, 3, a\}$

$\Rightarrow A \neq B$

eg:

$A = \{1, 2, 3, 4, 5\}$

$B = \{1, 2, 3, 5\}$

$\Rightarrow A \neq B$

Equivalent Sets: Two finite sets are said to be Equivalent if and only if $\underbrace{|A| = |B|}_{\text{i.e. same cardinalities}}$

eg. $A = \{1, 2, a, b, c\}$
$B = \{1, a, b, 2, c\}$
$\Rightarrow A = B$ & $|A| = |B|$ $\therefore$ A is equivalent to B

eg. $A = \{1, 2, 3, 4\}$
$B = \{1, 2, 3, a\}$
$\Rightarrow A \neq B$, but $|A| = |B|$, $\therefore$ A & B are equivalent

eg. $A = \{1, 2, 3, 4, 5\}$
$B = \{1, 2, 3, 5\}$
$\Rightarrow A \neq B$ & $|A| \neq |B|$
$\therefore$ A & B are not Equivalent as well

Note:

1. If two sets are equal, then they are Equivalent as well

2. If two sets are equivalent, then they may or may not be equal.

3. If two sets are not Equivalent, then they can never be equal

* **Note :—**

Let A & B are infinite sets.
A and B can be called Equivalent sets if and only if we can define a bijective function from set A to set B

If definition is not clear, then read it after functions.

Topic : Types of sets

Universal Set: It is a set of all elements associated with problem under consideration.

Generally denoted by U

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Topic : Types of sets

Subset & Superset:

A subset B → $A \subseteq B$

A superset B → $A \supseteq B$

A is a subset of B if and only if B is a superset of A

let A and B are any two sets, if Every element of set A is present in set B then we say that A is a subset of B (or B is a superset A)

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eg: $A = \{1, 2, a, b, c\}$
$B = \{1, a, b, 2, c\}$

Every element of A is included in B $\therefore$ $A \subseteq B$
Every element of B is included in A $\therefore$ $B \subseteq A$

if $A \subseteq B$ & $B \subseteq A$
then $A = B$
& Converse is also true

eg: $A = \{1, 2, 3, 4\}$
$B = \{1, 2, 3, a\}$

Neither $A \subseteq B$ nor $B \subseteq A$
i.e $A \nsubseteq B$ & $B \nsubseteq A$

$A \neq B$

eg: $A = \{1, 2, 3, 5\}$
$B = \{1, 2, 3, 4, 5\}$

$A \subseteq B$, but $B \nsubseteq A$ $\therefore$ $A \neq B$

Here A is a proper subset of B
i.e. $A \subset B$

Topic : Types of sets

Proper Subset: For a set A, any subset of set A Except set A itself will be proper subset of set A.

A proper subset of B

$A \subset B$

Equality Condⁿ is not allowed in Proper Subset.

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Note:

1. if $A \subseteq B$ & $B \subseteq A$, then $A = B$
2. if $A \subseteq B$ & $B \nsubseteq A$, then $A \subset B$
3. Every set is a subset of itself.
   i.e. $A \subseteq A$ is always true.
4. ϕ is a subset of Every set.
   i.e. $\phi \subseteq A$ is always true w.r.t. any set A.

eg. Let $A = \{a, b, c\}$

✓ Subsets of set A are $\Rightarrow$ $\{\ \}, \{a\}, \{b\}, \{c\}, \{a,b\}, \{a,c\}, \{b,c\}, \{a,b,c\}$

$\phi \subseteq A$ ✓ $\qquad$ $A \subseteq A$ ✓

Proper subsets of set A are $= \underbrace{\{\ \}, \{a\}, \{b\}, \{c\}, \{a,b\}, \{a,c\}, \{b,c\}}_{\text{All Subset}}$

Except → ✗ set itself is not allowed in Proper Subset.

Topic : Number of subsets of a set 'A' of cardinality 'n'

Let 'A' is a finite set of size = n, {i.e. $|A| = n$}

How many subsets of set A are possible = $\boxed{2^n}$

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$A = \{a, b, c\}$ $\quad |A| = 3$

Every subset of A will include 0 or more elements from set A, but no element from outside 'A'

Subsets of A, of size = 0	Subsets of A, of size = 1	Subsets of A, of size = 2	Subsets of A of size = 3
$\{\ \}$	$\{a\}, \{b\}, \{c\}$	$\{b,a\} = \{a,b\}, \{a,c\}, \{b,c\}$	$\{a,b,c\}$
Number of subsets of size = 0 $\Rightarrow {}^3C_0 = 1$	Number of subsets of size = 1 $\Rightarrow {}^3C_1 = 3$	Number of subsets of size = 2 $\Rightarrow {}^3C_2 = 3$	Number of subsets of size = 3 $\Rightarrow {}^3C_3 = 1$

Total no. of subsets of set A $= {}^3C_0 + {}^3C_1 + {}^3C_2 + {}^3C_3 = 2^3 = 8$

$= 1 + 3 + 3 + 1 = 8$

Note:

$$\boxed{{}^nC_0 + {}^nC_1 + {}^nC_2 + \cdots\cdots + {}^nC_n = 2^n}$$

Let $|A| = n$, then

Total no. of subsets of set A = Number of Subsets of size = 0 + Number of Subsets of size = 1 + ... + Number of Subsets of size = k $(k<n)$ + - - - + Number of Subsets of size = n

$$= {}^{n}C_0 + {}^{n}C_1 + {}^{n}C_2 + \cdots + {}^{n}C_k + \cdots + {}^{n}C_n$$

$$= 2^n$$

Let $|A| = n$, then

Total no. of subsets of set A = Number of Subsets of size = 0 + Number of Subsets of size = 1 + ... + Number of Subsets of size = K $(K<n)$ + - - - + Number of Subsets of size = n

$$= \underbrace{{}^nC_0 + {}^nC_1 + {}^nC_2 + \cdots + {}^nC_k}_{\text{Number of Subsets, such that size of each subset is at most } K \text{ (i.e. } \le K)} + \cdots + {}^nC_n = 2^n$$

Number of Subsets, such that size of each subset is at most K (i.e. $\le K$)

Number of Subsets of size exactly $= K$

Number of Subsets, such that size of each subset is at least K (i.e. $\ge K$)

Topic : Number of subsets of a set 'A' of cardinality 'n'

Let $|A| = n$,

① Total no. of subsets of set A $= {}^nC_0 + {}^nC_1 + \cdots + {}^nC_n = 2^n$

② Number of subsets of set A, such that size of each subset is exactly 'K' $= {}^nC_K$ {where $K \leq n$}

③ Number of subsets of set A, such that size of each subset is at most 'K' $= {}^nC_0 + {}^nC_1 + \cdots + {}^nC_K$

④ Number of subsets of set A, such that size of each subset is at least 'K' $= {}^nC_K + {}^nC_{K+1} + \cdots + {}^nC_n$

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Another Approach :-

$\boxed{\text{Number of Subsets}}$

Let $A = \{1, 2, 3, 4, \ldots\ldots, n\}$ i.e. $|A| = n$

$$\text{Number of Subsets of Set } A = \underbrace{2 * 2 * 2 * 2 * \cdots \cdot 2}_{\text{" n times 2 "}} = 2^n$$

2 Choiches.

$\downarrow$

it will be present in subset

(or)

it will not be present in subset

Q: Let $|A| = 5$,
How many subsets of set A are there of size $\underbrace{\text{at most}}$ '3'.

Solu^n: $\Rightarrow {}^5C_0 + {}^5C_1 + {}^5C_2 + {}^5C_3$

$\Rightarrow 1 + 5 + 10 + 10$

$\Rightarrow 26$

Q: Let $A = \{0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$

How many subsets of set A are there that contains exactly three odd numbers, and any no. of even numbers.

No. of odd numbers = 5

No. of Even numbers = 6

$\therefore$ No of subsets w.r.t. given condition $= {}^5C_3 * \{{}^6C_0 + {}^6C_1 + {}^6C_2 + {}^6C_3 + {}^6C_4 + {}^6C_5 + {}^6C_6\}$

$= 10 * 2^6$

$= 10 * 64$

$= 640$

Note :.

Let $|A| = n$.

then No. of proper subsets of set A $= \underbrace{2^n}_{\text{All subsets}} - \underbrace{1}_{\text{Except Set A itself}}$

* Concept w.r.t. number of Supersets :-

* Number of supersets of a given set will always be defined w.r.t. Universal set.

Q. Let $A = \{1, 2, 4, 5\}$ is a set derived w.r.t universal set $U = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$

How many supersets of set A are there ?

$U = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$

Number of supersets of set A $= 1 \times 1 \times 2 \times 1 \times 1 \times 2 \times 2 \times 2 \times 2 \times 2 = 2^6 = \underline{\underline{64}}$

These elements must be present. ∴ Only '1' choice for them

We know, $^{n}C_{r} = {}^{n}C_{n-r}$

No. of subsets of set A $= \underbrace{{}^{n}C_{0} + {}^{n}C_{1} + {}^{n}C_{2} + \dots + {}^{n}C_{k}}_{\text{No. of subsets of size = atmost 'k'}} + \dots\dots + \underbrace{{}^{n}C_{n-k} + {}^{n}C_{n-k+1} + \dots + {}^{n}C_{n-1} + {}^{n}C_{n}}_{\text{No. of subsets of size = at least 'n-k'}} = 2^{n}$

where $|A| = n$

No. of subsets of size = atmost 'k' = No. of subsets of size = at least 'n-k'

Q: Let $|A| = 5$,

How many subsets of set A are there of size at most '3' $\Rightarrow \underline{\underline{k=3}} \Rightarrow n-k = 5-3 = 2$

$\therefore$ No. of subsets of size at most '3' will be same as no. of subsets of size at least '2'

$$= {}^5C_2 + {}^5C_3 + {}^5C_4 + {}^5C_4$$

$$= 10 + 10 + 5 + 1$$

$$= 26$$

<u>Solu</u>

Solun:

$$\Rightarrow {}^5C_0 + {}^5C_1 + {}^5C_2 + {}^5C_3$$

$$\Rightarrow 1 + 5 + 10 + 10$$

$$\Rightarrow 26$$

Topic : Power Set

Power Set: Let 'A' is any finite set, then power set of set 'A' is a set of all subsets of set A.

Power set of set A is denoted by $P(A)$ or 2^A

$$\boxed{P(A) = \text{Set of all subsets of set } A}$$

$$P(A) = \{ \underbrace{\qquad\qquad\qquad\qquad}_{\text{Elements will be all subsets of set A}} \}$$

Set

Slide

Q: Let $A = \{a, b, c\}$

then

$P(A) = ?$

Soln

$$P(A) = \underset{\text{Set}}{\{}\underbrace{\{\,\}, \{a\}, \{b\}, \{c\}, \{a,b\}, \{a,c\}, \{b,c\}, \{a,b,c\}}_{\text{elements are all subsets of set } A.}\}$$

Note:-

① $P(A)$ is also a set, and we know that ϕ is a subset of Every set

$\therefore \phi \subseteq P(A)$ is true for any set A.

② ϕ is always an element of $P(A)$, for any set A.

i.e, $\phi \in P(A)$ is true for any set A

③ For any set A

$A \in P(A)$ is always true

Topic : Cardinality of Power Set

Cardinality of $P(A)$ $= |P(A)| =$ No. of elements in $P(A)$

$=$ Total no. of subsets of set A.

$= 2^{|A|}$

$$\boxed{|P(A)| = 2^{|A|}}$$

Slide

H.W.

Q:- Let $\varnothing$ be an Empty set, then

$$|P(P(\varnothing))| = ?$$

(a) 0

(b) 1

(c) 2

(d) Can not be defined.

H.W.

Q: Let $A = \{1, 2, 3, \{1,2\}, \{\{2,3\}\}\}$

Which of the following is/are true

(a) $1 \in A$

(b) $\{1\} \in A$

(c) $1 \subseteq A$

(d) $\{1\} \subseteq A$

(e) $\{1,2\} \in A$

(f) $\{1,2\} \subseteq A$

(g) $\{\{1,2\}\} \in A$

(h) $\{\{1,2\}\} \subseteq A$

(i) $\{2,3 \in A$

(j) $\{2,3\} \subseteq A$

(k) $\{\{2,3\}\} \in A$

(l) $\{\{2,3\}\} \subseteq A$

(m) $\{\{\{2,3\}\}\} \subseteq A$

(n) $\emptyset \in A$

(x) $\emptyset \subseteq A$

(y) $\emptyset \in P(A)$

(z) $\emptyset \subseteq P(A)$

H.W.

Q2. For a set A, the power set of A is denoted by 2^A. If $A = \{5, \{6\}, \{7\}\}$ which of the following options are true?

1. $\emptyset \in 2^A$
2. $\emptyset \subseteq 2^A$
3. $\{5, \{6\}\} \in 2^A$
4. $\{5, \{6\}\} \subseteq 2^A$

Slide

H.W.

Q1. Let $P(S)$ denote the power set of a set S. Which of the following is always true?

A. $P(P(S)) = P(S)$

B. $P(S) \cap P(P(S)) = \{\emptyset\}$

C. $P(S) \cap S = P(S)$

D. $S \notin P(S)$

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2 mins Summary

Topic Sets and representation of sets

Topic Different types of sets

Topic Number of subsets of a given set

Topic Power sets

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THANK - YOU

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